Amazon Sales Dashboard

Project Overview

This project focuses on analyzing Amazon sales data using Power BI to identify sales trends, customer behavior, product performance, and business insights. The dataset contains more than **100,000 rows** and **18 columns**, providing detailed information about orders, customers, products, sales, payment methods, returns, and other business-related metrics.

Tools Used

- Power BI
- Microsoft Excel
- Power Query
- DAX

Project Workflow

1. Understanding the Problem

The objective of this project was to analyze Amazon sales data and generate meaningful business insights that can support data-driven decision-making. The focus was on understanding sales performance, customer purchasing patterns, product demand, and overall business growth.

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2. Understanding the Data

The dataset contains over **100,000 records** and **18 attributes** describing various aspects of Amazon sales transactions.

The following checks were performed:

- Examined column names and their business meaning
- Identified data types (Text, Number, Date, Currency)
- Checked for missing values
- Identified duplicate records
- Understood relationships between different fields
- Validated data consistency

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3. Data Cleaning

Data cleaning was performed to improve data quality and ensure accurate analysis.

Tasks completed:

- Removed duplicate records
- Corrected data types
- Handled missing values
- Fixed inconsistent data entries
- Removed unnecessary columns
- Checked and handled outliers where required
- Standardized data for analysis

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4. Data Transformation

The cleaned data was transformed using Power Query to make it suitable for reporting and analysis.

Transformation activities:

- Created calculated columns
- Formatted date fields
- Generated business metrics
- Organized and structured data tables
- Prepared data model for dashboard creation

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5. Exploratory Data Analysis (EDA)

Exploratory analysis was conducted to discover patterns, trends, and business opportunities.

Analysis included:

- Average Sales Analysis
- Maximum Sales Analysis
- Minimum Sales Analysis
- Product Performance Analysis
- Customer Purchase Behavior
- Revenue Trend Analysis
- Order Distribution Analysis

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6. Data Visualization

Interactive visualizations were created using Power BI to present insights in a simple and effective manner.

Dashboard components include:

- KPI Cards
- Bar Charts
- Line Charts
- Pie Charts

- Filters and Slicers
- Trend Analysis Visuals

These visualizations help stakeholders quickly understand business performance.

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7. Insights & Interpretation

The dashboard helped identify:

- Top-performing products
- High-value customers
- Sales trends and patterns
- Most preferred payment methods
- Returned and cancelled order behavior
- Revenue contribution across categories

These insights can support business strategy and operational improvements.

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8. Reporting

A comprehensive Power BI dashboard was developed to summarize key findings and present business insights in an interactive format.

The report enables stakeholders to:

- Monitor sales performance
- Track customer behavior
- Analyze product demand
- Support strategic decision-making

Dashboard Screenshot

Amazon Sales Dashboard

(Add your dashboard screenshot here)

Key Features

- Interactive Dashboard
- Sales Performance Tracking
- Customer Analysis
- Product Analysis
- Revenue Insights
- Business KPI Monitoring
- Data-Driven Decision Support

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